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Preparation and Properties of 5-Fluorouracil-Loaded Chitosan Microspheres for the Intranasal Administration

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ABSTRACT

Objective To study the preparation technique of 5-fluorouracil and the release characteristic of 5-fluorouracil-loaded chitosan microspheres for the intranasal administration.

Methods 5-fluorouracil-loaded chitosan microspheres were prepared by emulsion chemical cross-link technique. The orthogonal experimental design was used to optimize the preparation procedure. Dynamic dialysis method was applied to determine the release characteristic of microspheres in vitro and its influencing factors. Swelling behavior was expressed by swelling ratio. The degree of mucoadhesion was investigated by determining the mucociliary transport rate(MTR) of the microparticle across a frog palate.

Results Microspheres with a good shape and narrow size distribution were prepared. The average diameter was 43 ± 4 µm. The drug loading was (38.5 ± 1.0) %. The entrapment efficiency was (79.0 ± 1.8) %. The drug release profile in vitro could be described by Higuichi eqution as $Q = 0.1035t^{1/2} + 0.0284$ ($r = 0.9965$). Chitosan had good mucoadhesive property and caused a significant reduction in MTR($P < 0.01$).

Conclusion The optimized technique has a good reproducibility and a high entrapment efficiency, so it could be used to prepare 5-fluorouracil-loaded chitosan microspheres for the intranasal administration.Chitosan is a good material for nasal preparation and has prospective development in the pharmaceutical field.

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Central nervous system (CNS) diseases include brain tumors, central nervous system infections, chronic diseases, drug addiction, epilepsy, cyclical migraine, neurodegenerative diseases, schizophrenia and stroke. There are a great number of patients that suffer from these diseases, thus CNS treatment is needed. But due to the existence of blood-brain barrier (BBB) [1–3], more than 95 % of the central nervous system drugs can not pass through the BBB. Therefore, how to transmit the CNS drugs in a non-invasive way into the brain has become popular for pharmaceutical workers. It has been reported that there are many approaches to deliver drugs into the brain, and the nasal administration which is through the olfactory mucosa directed to the brain is a very active field over the last decade. As the nasal mucosa is anatomically and physiologically connected to the brain, it is expected to transmit the drugs through the nasal mucosa to bypass the BBB directly and target to brain [4].

There has been reports about fluorouracil (5-fluorouracil,5-Fu) solution that can be directly targeted to the brain [5–7] by nasal administration to treat brain tumors. In this paper, chitosan (CH) with bioadhesive and absorption-promoting properties was used as carrier material. Fluorouracil was used as model drug to prepare nasal brain-targeted chitosan microspheres. Pharmacokinetic studies will be published separately.

* Wanqing Li mainly contributed to the study

Materials and Methods

Drugs and Reagents

Fluorouracil (Shandong Qilu Pharmaceutical Factory); Chitosan (Qingdao Haihui Company, 100 mesh)

Deacetylation degree of 75 % (I), deacetylation degree of 85 % (II), deacetylation degree of 90 % (III); liquid paraffin (chemical plant in Tianjin), chemically pure; formaldehyde (Beijing Chemical Reagent Company), analytically pure; the rest reagents are all analytically pure.

Instruments and equipment

Electromagnetic mixer (Shanghai Caoxing Radio Components Factory); Electromagnetic mixer (Changzhou Guohua Electric Co., Ltd.); Optical microscope (Beijing Precision Instrument Factory); JSM-5600 Scanning Electron Microscope (Japan JEOL Corporation); Dialysis membrane (molecular weight cutoff 10,000); 752 UV spectrophotometer (Shanghai Precision Science Instrument Co., Ltd.).

Preparation of Chitosan Microspheres

Chitosan solutions with different concentrations were prepared with 5 % acetic acid as aqueous phase. Then put proper amount of liquid paraffin and emulsifier into a 100 ml beaker and mix it as oil phase. While agitating, the water phase is dripped into the oil phase by a syringe to prepare the w/o type emulsion. After 30 min' emulsification, cross-linking agent is added to solidify it. Then stand it for a while and pour the oil phase out. The precipitate is then washed by petroleum ether, dehydrated by absolute alcohol. After vacuum drying, blank microspheres were prepared.

The drug loaded microspheres were obtained as follows. 5-Fu was first dissolved in 5 % acetic acid, then chitosan was added to dissolve and mix as water phase. Subsequently, the preparation method was identical to that of blank microspheres.

Particle size and morphology of microspheres

The sphericity and dispersion of dry microspheres were observed under an optical microscope with scale. Measure 500 particles in a selected area and calculate the average particle size.

The dry microspheres were uniformly dispersed on the sample platform. Then coat gold after fixing it with a conductive adhesive. The surface morphology of the microspheres was observed by scanning electron microscopy(SEC).

Determination of 5-Fu by Ultraviolet Spectrophotometry

UV scanning

An appropriate amount of 5-Fu and chitosan were dissolved in 0.1 mol/ml HCI respectively. Take HCl as blank control, ultraviolet scanning was operated ranging from 200 to 400 nm. Compare with the scan spectrum, and select the best absorption wavelength.

Plotting of standard curves

Take about 10 mg of 5-Fu raw powder and weigh it accurately, put it in a 50 ml volumetric bottle. Add 0.1 mol/L HCl to dissolve and fix it to the scale. Then dissolve 0.125, 0.25, 0.5, 1.0, 2.0, 5.0, 10 ml into 50 ml volumetric flasks respectively, add 0.1 mol/L HCl to set the scale and measure the absorbance. At the selected wavelength, the optical density was determined by using a UV-260 UV spectrophotometer to calculate the standard curve.

Determination of recovery rate

Absorb a proper amount of 5-Fu standard solution respectively and accurately, add it into a set of blank chitosan microspheres of high, medium and low content.Add 0.1NHCL50 ml, reflux it in water bath for 3 h at 60℃ to dissolve the microspheres completely, then calculate its content with the standard curve and compared with the amount added to calculate the recovery rate.

Determination of drug-loading content and encapsulation efficiency of microspheres

Take about 50 mg of microsphere, weigh it accurately and place it in a 100 ml flask. Then add 0.1 mol/L HCL 50 ml, reflux it in a water bath about 3 h at 60℃, make the microsphere dissolve completely. Cool down, set the scale and filter it. Take the 1 ml of filtrate and dilute it to 10 ml, measure the optical density at 266 nm. The concentration was determined by standard curve. The drug-loading content and encapsulation efficiency were calculated according to the following formula:

Drug-loading content = ρ52Fu × 0.01 × 100/m (the unit of ρ is mg/L)

Encapsulation efficiency = weight of drug in the microsphere/ total dose administered

In vitro release

About 50 mg of 5-Fu chitosan microspheres were placed in a dialysis bag and 2 ml of PBS at pH 6.5 was added, then clamped it with dialysis clamp and placed it in 97 ml of PBS at pH 6.5. Dialyze it dynamicly at 37 ± 1℃. Take 1.0 ml from external solution for a certain period of time, also supplement 1.0 ml PBS meanwhile. Then dilute the sample liquid with 0.1 mol/L HCl to 5 ml. Measure the UV optical density at 266 nm, calculate the drug concentration from the standard curve equation, and calculate the cumulative release of the drug and the cumulative release percentage.

Determination of swelling ratio (Rs) of chitosan microspheres

The swelling ratio is expressed by the water absorption capacity of microspheres [8, 9]. The dried microspheres (about 50 mg) were weighed, then placed in room temperature PBS (pH = 6.5) and taken at different times. Wipe the water on the surface of the swelling microspheres with filter paper. Weigh its weight quickly with an electronic balance, and calculate Rs as following equation:

$$Rs = \frac{\text{Wet weight} - \text{Dry weight}}{\text{Dry weight}} \times 100$$

Determination of mucoadhesion

The mucosal adhesion of the microspheres was evaluated by measuring the mucociliary transport rate [10–12].

Fix the frog onto the frog board, pull the lower jaw down with a thick cotton thread through the lower jaw and fix the thread to the nail between the hind legs, and then fasten the upper jaw with

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pins, so that the maxillary mucosal surface was fully exposed. Use Ringer's solution to moisturize the mucosal surface.

The carbon particles were placed on the mucosa of the anterior margin of the 2 eyes, and the carbon particles were observed to move toward the esophagus due to the ciliary movement of the mucosal epithelium. The time required for the particles to move 1.5 cm was measured with a stopwatch. And the mean mucociliary transport rate was taken as the base mucociliary transport rate of frogs, and then the blank microspheres or drug-loaded microspheres were placed on the mucosa to calculate the mucociliary transport rate of the microspheres The greater the mucosal adhesion of the microspheres, the smaller the mucociliary transport rate.

Results

Determination of 5-Fu content

The UV scan shows that the maximum absorption rate of 5-Fu is at 266 nm, while the absorption of chitosan is very small, and the chitosan does not interfere with the measurement. Thus the amount of 5-Fu can be determined by ultraviolet spectrophotometry. So 266 nm is the selected wavelength for the determination of 5-Fu, and make the standard curve equation at such wavelength, the equation is:

D(266 nm) = 0.0528ρ52Fu + 0.007(the unit of ρ is mg/L)

Dissolve the microspheres of low, medium and high content we mentioned in 1.5.3 and calculate their contents through standard curve, and compare with the amounts added to calculate the recovery rates. The average recovery rates were (102.6 ± 1.3) %, N = 9.

Preparation of 5-Fu chitosan microspheres

Selection of chitosan species

The deacetylation degree of chitosan is an important factor influencing the preparation of microspheres. With the deacetylation degree of chitosan increases, viscosity of solution decreases, the size of microspheres decreases rapidly, the dispersion is getting worse, and the color is darker. Therefore, we chose the chitosan with 85 % deacetylation degree in this experiment.

Selection of dispersion medium

The organic phase dispersion medium has a great influence on the particle size and the formation of microspheres. When toluene is used as a dispersing medium, microspheres become spherical poorly, the shape of particle is irregular and the adhesion is serious. While castor oil is at the same amount of emulsifier, the sphere is not as good as liquid paraffin, and the spherule is not round enough. It is best to use liquid paraffin as the organic dispersion medium for the reaction.

Selection of curing agent type

Preliminary experiments and literature show that the curing ability of formaldehyde is much weaker than that of glutaraldehyde. Under the same preparation conditions, 2 batches of microspheres were prepared respectively with formaldehyde and glutaraldehyde as the curing agent, digest with 0.1 mol/L HCl. Microspheres prepared with formaldehyde as curing agent, 60℃ water bath, can be completely digested in 3 h. And the microspheres prepared with glutaraldehyde as curing agent were still intact granules under the same condition. The duration of staying in the nasal cavity is limited due to the nasal administration of the drug. Furthermore, the aim of this study is to prepare bioadhesive microspheres instead of slow-release microspheres. Therefore, formaldehyde is chosen as the curing agent, so that the medicine can be released to the maximum extent in a limited time.

▶ **Table 1** Orthogonal factors and levels.

chitosan concentration	stirred rate(rpm)	mass ratio of 5-Fu/CH	cross-linking time
1.5 %	700	1:2	1 h
2 %	1000	2:3	2 h
3 %	1500	1:1	3 h

Optimization of the preparation process by orthogonal design

Based on the reference literature and the results of the single factor investigation, 4 factors that influences the particle size, distribution and drug loading content of microspheres were taken into consideration: the concentration of chitosan, stirred rate, mass ratio of 5-Fu/CH and cross-linking time. 3 levels for each factor, as shown in ▶ **Table 1**. The experiment was arranged according to L9(34) orthogonal design table.

The statistical methods [13] of this study refer to the literature. 3 important items controlling microspheres quality were investigated: particle size distribution, microsphere drug-loading content, encapsulation efficiency. In the orthogonal design experiment, the ideal function (DF) was introduced to evaluate the preparation process of microspheres synthetically.

From the orthogonal design, the order of importance of 4 factors is: chitosan concentration > 5-Fu/CH mass ratio > stirred rate > curing time. Chitosan concentration, stirring speed and curing time mainly affected the diameter distribution percentage, while chitosan concentration and 5-Fu/CH mass ratio affected the loading content and encapsulation efficiency of microspheres. By visual analysis, the optimized prescription of microspheres was $A_3B_1C_3D_2$, that is, chitosan concentration is 3 %, stirred rate is 1000 rpm, 5-Fu/CH ratio is 1:1, cross-linking time is 3 h. Specific operations are as follows: take 0.27 grams of 5-Fu in 9 ml 5 % acetic acid solution, completely dissolved, and then add 0.27 g chitosan to make it fully swollen. Pour the liquid 6 ml into the 60 ml liquid paraffin containing 2 % span80, emulsify 30 min at 1000 rpm, then add formaldehyde 3 ml, solidify 3 h. Let stand, dump the oil layer, take it washed 3 times each with petroleum ether and absolute ethanol. Filter, vacuum dry, then finish.

4 batches of drug-loaded microspheres were prepared by orthogonal experimental optimized process. The size distribution of 20 ~ 60 μm microsphere accounts for 71.76 ± 2.17 %, the loading content of microspheres was 29.2 ± 0.9 %, and the encapsulation efficiency was 61.3 ± 1.68 %. The results showed that the preparation process of microspheres was stable and well-reproducible.

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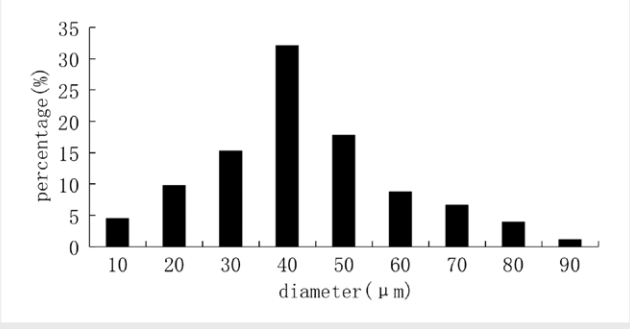
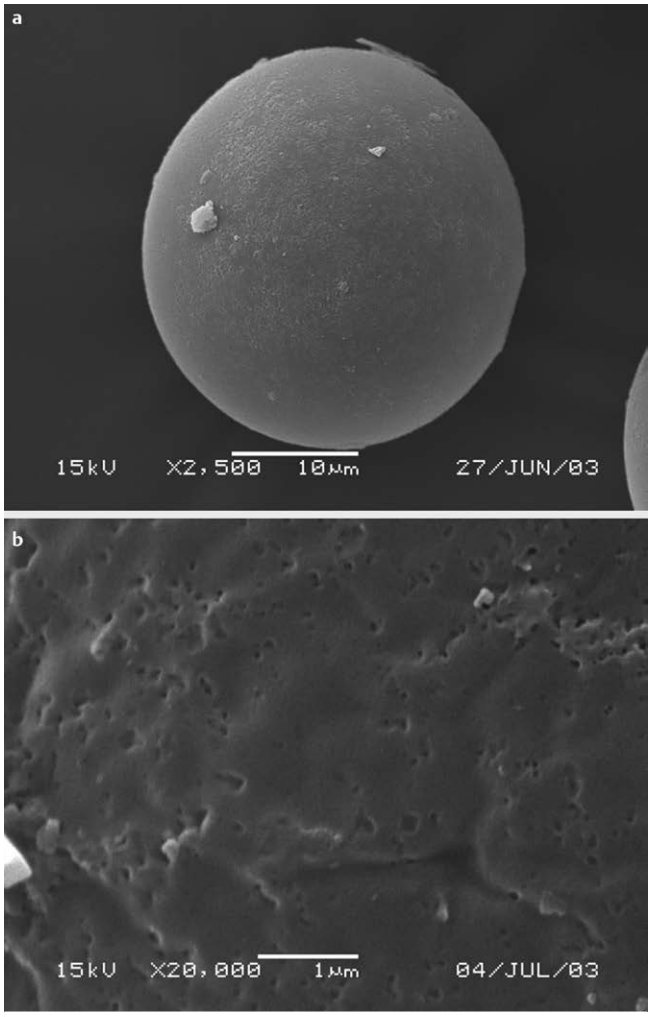


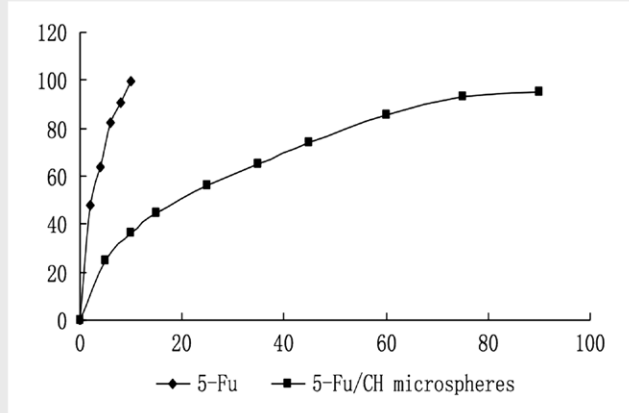
▶ **Fig. 1** Size and distribution of 5-Fu loaded chitosan microspheres.



▶ **Fig. 2** SEM photography of 5-Fu-loaded chitosan microspheres (**a**) are rough, spherical rounding. And there are few drug microcrystals, indicating that the drug has been encapsulated. The fine structure (**b**) of the microspheres is similar to the fine structure of the blank microspheres, and also has tiny holes formed after the dissolved water is discharged.

Drug loading content and encapsulation efficiency of microspheres

As 5-Fu has suitable solubility in the oil phase [14], it can disperse into the oil phase, and the content of the drug in the microspheres will be greatly reduced. In order to improve the drug loading content and encapsulation efficiency of 5-Fu chitosan microspheres, the saturated oil phase method can be used. That is, the liquid paraffin is first saturated with 5-Fu, then use the upper clear layer of the saturated liquid paraffin as the oil phase, prepare 4 batches of drug-loaded microspheres in accordance with the orthogonal experiment optimization process. The size distribution of 20~60 μm microsphere accounts for 73.9 ± 2.4 %, the drug loading content of microspheres was 38.5 ± 1.0 %, and the encapsulation efficiency was 79.0 ± 1.8 %. Compared with the previous values, the drug loading content and encapsulation efficiency of microspheres were significantly improved.



▶ **Fig. 3** Drug release from chitosan microspheres in vitro.

Particle size and morphology of microspheres

Observed by light microscopy, the microspheres were spherical rounding, well dispersed and uniform in size. ▶ **Fig. 1** is a histogram of particle size distribution, in which 20~60 μm microsphere accounts for 73.93 % of the total, and the average diameter of the microsphere is 43 ± 4 μm.

▶ **Fig. 2** shows the SEM images of the drug-loaded microspheres. The microspheres (**a**) are rough, spherical rounding. And there are few drug microcrystals, indicating that the drug has been encapsulated. The fine structure (**b**) of the microspheres is similar to the fine structure of the blank microspheres, and also has tiny holes formed after the dissolved water is discharged.

Drug release of microspheres in vitro

The in vitro release of 5-Fu-loaded chitosan microspheres is shown in ▶ **Fig. 5**. And the release curve can be divided into 2 parts. The front part of the drug release curve steep rapidly,15 min when the cumulative release of the amount that is up to 40 %, the sudden release effect was significant. The posterior part of the curve is flat and the release is slow. The release behavior is fitted by zero, first-order and Higuichi equations, and the linear equation is obtained by regression:

$Q = 0.0095t + 0.2335$, $r = 0.9382$(zero order)

$\ln(1-Q) = -0.0329t - 0.0328$, $r = 0.9948$ (first order)

$Q = 0.1035t^{1/2} + 0.0284$, $r = 0.9965$ (Higuichi equation)

Therefore, the fitting effect of release in vitro is best fitted by Higuchi equation (▶ **Fig. 3**).

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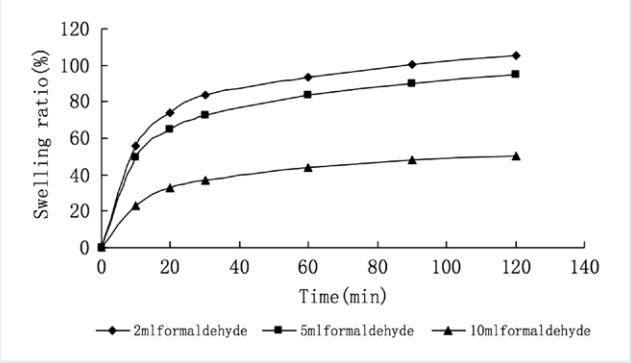
The determination of swelling degree of the microspheres

Chitosan microsphere is a kind of hydrophilic gel microsphere. When injected into the nasal cavity, chitosan can rapidly absorb the surrounding water and swell to form in-situ gel, so it has good biological adhesion. In order to investigate the relationship between swelling and release of microspheres, chitosan microspheres with different curing agents were prepared. The swelling curves of microspheres were shown in ▶ **Fig. 4**, and the release curves of microspheres were shown in ▶ **Fig. 5**. The figure shows that the amount of curing agent is negatively correlated with the swelling of microspheres and the in vitro release. This is because the greater the amount of curing agent, the greater the cross-linking density, the greater the strength of the microspheres, the smaller the degree of swelling of the microspheres, while the smaller the microchannels are, and it'll be harder for water to get in and get out, the greater the resistance of drug molecules to release and spread, the slower the drug release rate is.

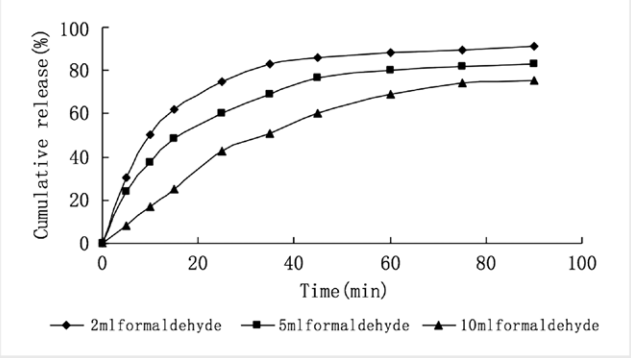
From the swelling curve and release curve, it can be seen that the 2 curves are very similar, indicating the consistency between these 2. In the early stage of release, microspheres rapidly absorb water through pores and swell rapidly, resulting in rapid release of microspheres. Release behavior follows the swelling control mechanism. The faster the swelling, the quicker the drug release. And when the swelling reaches equilibrium, the drug can only slowly diffuse through the skeleton or release slowly through the degradation of chitosan, then release rate will be slowed down and release follows the diffusion control mechanism.

Determination of mucoadhesion

The mucosal adhesion of blank microspheres and drug-loaded microspheres was examined. 6 frogs were used for one single sample in the t test.Both the blank microspheres and the drug-loaded microspheres reduced the mucociliary transport rate significantly ($P<0.01$), so chitosan was a good biosorbent (▶ **Table 2**).



▶ **Fig. 4** Swelling behaviour of chitosan microspheres prepared with different amounts of cross-linker.



▶ **Fig. 5** Drug release from chitosan microspheres prepared with different amounts of cross-linker.

▶ **Table 2** Reduction of MTR in empty and 5-Fu-loaded microspheres ($\bar{x} \pm s$, n = 6).

Test microparticles	Basal MTR (mm/min)	MTR (mm/min)
5-Fu-loaded microspheres	8.45 ± 1.29	3.07 ± 0.56 **
empty microspheres	8.06 ± 1.37	2.66 ± 0.44 **
** $P<0.01$		

Discussion

Chitosan which is made of deacetylated chitin is a kind of macromolecule amino polysaccharide, and is the only natural presence of alkaline polysaccharides, the molecule contains a large number of free amino and hydroxyl, with which mucus glycoprotein form hydrogen bonds and thus has a good biosorption, while the molecular structure is of positive charge, and can interact with negatively charged mucosa to open the intercellular tight junction between the cells to promote drug intercellular transportation, it can enhance the ability of drug to penetrate membrane, so is a good mucosal absorption enhancer. Chitosan can also inhibit tumor cells, and the anti-cancer effect will be enhanced when combined with 5-FU. 5-FU based NDDS with targeting capability would not only reduce the systemic side effects, but also provide an effective and safe therapy with a reduced dose and duration of therapy [15]. Therefore, we used 5-FU to treat brain tumors by nasal administration, which can be directly targeted to the brain.

5-Fu is an acidic, water soluble drug, and is an antineoplastic agent of wide use for the treatment of solid tumors in clinical chemotherapy [16], but soluble in hot acid solution. Chitosan is soluble in diluted acid solution, but in acidic conditions, degradation occurs, so we choose acetic acid which is less acidic. As the acetic acid aqueous solution of chitosan, especially when in high concentration, is relatively thick, and H^+ and the second-NH2 binding acidity decreased, 5-Fu is not easy to dissolve but turn into powder mixed in the solution. With the proportion of chitosan increased, the turbidity gradually increased. Therefore, first prepare higher concentration of 5-Fu acetic acid aqueous solution, and then add the appropriate amount of chitosan, and heated and stirred to prepare a clear aqueous solution of 5-Fu chitosan.

Drug loading within the polymer matrix depends upon the release, however, drug release through microspheres depend on the particle size [17] and the particle size of the nasal mucosal microspheres has a great influence on the distribution of drugs in the nasal cavity. When the particle size < 10 µm, the microspheres will enter the bronchial with the airflow, and when the particle size is

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large, the microspheres are mainly distributed in the non-cilia area of the nasal cavity. Therefore, the particle size of the nasal mucosal microspheres is preferably in the range of 40 to 60 m. The average particle size of 5-Fu chitosan microspheres prepared in this paper is 43.2 μm, which meets this requirement.

Burst effect is conducive to the nasal absorption of drugs, drugs can be injected into the nasal cavity in a short time to achieve a higher concentration, while reducing lag phenomenon. (▶ **Fig. 2**) in the early release, PBS can be quickly pass through the micropores into the inside of micro-ball, and dissolve the drug to form a higher, or even saturated concentration, according to the difference of concentration, drug molecules in the the micro-balls disperse from the micropores quickly to form bursts. The chitosan is easy to swell, so that the pore diameter decreases or even disappears, so the pore channel is closed, and the subsequent drug release can only slowly spread through the skeleton, and chitosan forms a denser matrix, making it difficult for 5-FU particles to diffuse through the viscous gel layer of the matrix, which slowed the release of 5-FU [18]. Another reason for the formation of bursts is the microscopic addition of drugs to the surface of the microspheres.

To explore the drug release characteristics, it is important to study the swelling of microspheres. The polymer structure has an effect on the nature and extent of interaction between the solvent molecules and the polymer chains, because the porosity of the microspheres and the nature of hydrophilic groups of the polymer make an important impact [19].

Nasal mucosal surface is a layer of pseudostratified ciliated columnar epithelium [20], and there are a large number of goblet cells that secrete, the mucus often covers in the cilia. Under normal circumstances, cilia swing simultaneously to remove the nasal foreign body, cilia removal can cause the retention time of the drug in nasal cavity become short, and efficacy is affected, thus the evaluation of the adhesion of the preparation is of great significance. Increasing the adhesion of the preparation can prolong the action time of the drug in the nasal cavity and enhance the absorption of the drug. The stronger the adhesion of the preparation is, the slower the mucociliary transport rate is, and the more the decrease is compared with the basal mucociliary transport rate, the mucociliary transport rate of the preparation can be statistically compared to the basal mucociliary transport rate of the carbon particles (T test), and the adhesion of the nasal preparation was evaluated.

Conclusion

In this study, by applying emulsion chemical cross-linking technique and saturated oil phase method, the nasal mucosal drug fluorouracil chitosan microspheres were prepared, this technique is simple and easy to promote to large-scale production. The swelling speed of the microspheres is high, the swelling rate is high, the release process of the drug is very similar to the swelling process, and the drug concentration can be obtained in a short time after the administration, which is compatible with the short residence time of the nasal cavity. Chitosan microspheres have a good bioadhesion that can significantly prolong the residence time of the drug in the nasal cavity, thereby increasing the bioavailability of the drug.

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Conflict of Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled, “Preparation and properties of 5-fluorouracil-laoded chitosan microspheres for the intranasal administration”.

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